
Dengue outbreak prediction model for urban Colombo using meteorological data

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Abstract: Dengue is a viral borne disease with complex transmission dynamics. Disease outbreak can exert an increasing pressure on the health system with high mortality. Understanding and predicting the outbreaks of dengue transmission is vital in controlling the spread. Mathematical models have become important tool in predicting the dynamics of dengue. Due to the complexity of the disease, general time series models do not describe the impact of the external parameters. In this work, we propose a generalised linear regression model to understand the dynamics of the dengue disease and predict the future outbreaks. To moderate the model, cross-correlation between reported dengue cases and climatic factors were identified using Pearson cross-correlation formula. Then threshold value was defined based on reported data in order to identify minimum risk level for the states of dengue outbreaks. Further, obtained results were compared.

Keywords: dengue; vector density; climate factors; cross-correlation; Pearson correlation formula; time lag; generalised linear regression formula; disease outbreak; threshold; prediction model.

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1 Introduction

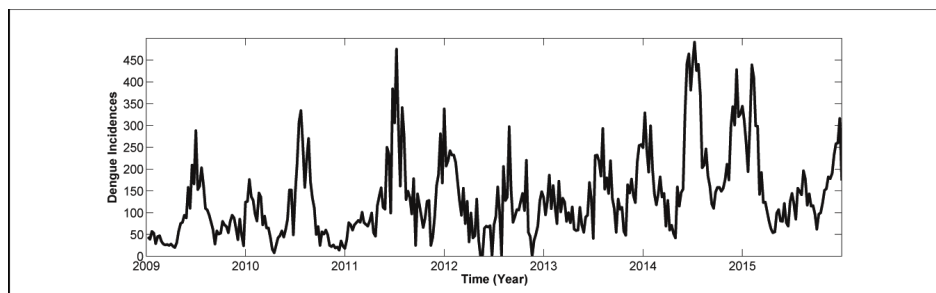
Dengue is a viral infection which is transmitted by bites of infected female *Aedes Aegypti* or *Aedes Albopictus* mosquitoes. Number of dengue cases has rapidly spread during the past few decades and it has become one of the major public health issues in tropical and subtropical regions in the world (Messina et al., 2014; World Health Organization, 2009). According to (World Health Organization, 2009), dengue is endemic in more than 100 countries and 0.4–1.3 million cases are reported annually.

Sri Lanka is a tropical country which has been affected by dengue since 1962 and the infection has now gained the status of an endemic disease. The country experienced its first outbreak during 1965–1966 (Sirisena and Noordeen, 2014). Since then, several dengue outbreaks occurred and the worst was reported in 2017 with 186,101 cases. Colombo Municipal Council (CMC) area is the most urbanised area in the country and 25% of the annual dengue fever cases of the country were reported from the CMC area (Wickramaarachchi and Perera, 2017). Figure 1 illustrates dengue infection trend of CMC area from 2009 to 2015.

The disease is caused by any one of the four serotypes of dengue virus, namely DENV 1, DENV 2, DENV 3 and DENV 4 which are serologically related and infection of one serotype of dengue virus does not provide cross immunity against the other three serotypes (Gubler, 1998). Thus, persons can be infected with multiple serotypes during their lifetime and persons with repeated infection have a risk of developing severe dengue conditions known as dengue hemorrhagic fever (DHF). Since specific drugs are not currently available and developing a vaccine for the disease is at an early stage of experimentation (Bhatia et al., 2013; Gubler, 1998; World Health Organization, 2011), the main prevention strategy is

vector controlling (World Health Organization, 2009). Day by day vectors are becoming more resistant to the commonly used insecticides and control strategies (Hemingway and Ranson, 2000). Hence, the number of reported cases increase dramatically. Vector controlling imposes a significant taxation on the economy of the country. Therefore, it is necessary to understand the dynamics of the dengue spread to minimise the cost. It may be possible to find more efficient control strategies, once the outbreaks of the epidemic is predicted for next few months. The activities to mitigate and control the outbreak situation currently based on the reported weekly average dengue incidence for the last 5 years and the method might yield little benefit. Accordingly, understanding the spreading pattern and predicting the outbreak of the disease is vital in controlling of dengue outbreaks.

Figure 1 Reported dengue cases from 2009 January to 2015 December, Colombo Municipal Council area



Since dengue is a vector born disease, the spreading trend of the disease depends on the number of infected mosquitoes. Accordingly, number of adult mosquitoes in an area is a major factor which affects the spreading trend of dengue. We denote it as the *vector density* and calculating this density is an infeasible task. Mosquito density is a highly sensitive parameter which heavily depends upon climatic factors such as rainfall and temperature (Erickson et al., 2012). For example, several previous works have examined the influence of climate factors on dengue distribution in different regions of Sri Lanka (Arunachalam et al., 2010; Ehelepola et al., 2015; Goto et al., 2013; Pathirana et al., 2009; Wickramaarachchi and Perera, 2017, 2018). The country is influenced by two monsoon seasons and the disease occurs every year within or soon after the monsoon seasons because rainfall contributes to generate vector-breeding sites. Some studies (Pathirana et al., 2009; Wickramaarachchi and Perera, 2018) concluded that reported dengue incidence have strong correlation with rainfall in Colombo with different time lags. Moreover, temperature of Sri Lanka varies from 17° to 35° which is ideal for vector biology, vector-virus interactions and rapid replication of the virus. Furthermore, unplanned urban development and land-use have blocked the flow of rainwater in urban areas. As a consequence the collection of rainwater in low wetlands provide ideal breeding sites for mosquitoes.

Due to the interaction of all these factors, it is extremely difficult to predict dengue outbreaks using classical mathematical and statistical models. Generally, time series model may be used to identify the seasonal patterns and/or statistical dependencies within the dataset. However, general time series models do not describe external parameters impacts. Hence, fail to provide best-fitting predictive model if there is significant external impact (Lu et al., 2009). Moreover, all these classical models are mainly based on parameters insensitive to time or space. Several statistical models can be found in literature which

attempted to investigate dengue outbreaks and its dynamics with respect to climate variation using regression-based methods (Carrington et al., 2013; Chen et al., 2010; Goto et al., 2013). Most of these models take one parameter at each time (Goto et al., 2013; Hales et al., 2002). Although some studies used multiple regression models, less attention has been paid to time lag between meteorological data and dengue incidence (Siriyaasatien et al., 2016). Furthermore, lag time between meteorological factors and dengue changes with time for a particular area. For example, study (Pathirana et al., 2009) concluded a 3–4 weeks lag for the period 2000 to 2004 and (Wickramaarachchi and Perera, 2018) concluded an 8 weeks lag time for the time period of 2006 to 2011 between the precipitation and dengue for Colombo. In this study, we examine how to use multiple climate factors and reported dengue data to develop an real time climate driven warning system that is able to predict the spreading pattern of the disease for next time period. Since generalised regression models are particularly interesting when there are time dependence between variables and possible to model a temporal structure with more reliable predictions (Gharbi et al., 2011), we adopt the generalised linear regression model as the basis for the study and moderate the model to incorporate lag time for meteorological data. Consequently, we yield a formula to predict dengue outbreaks in urban Colombo.

We organise the paper as follows. First, we select the appropriate model in Section 2.1. Then discuss about the CMC area in detail in Section 2.2. Data analysis methods are discuss in Section 2.3. This is followed by the results we obtained from data analysis and model simulations, that will be described in Section 3. Our concluding remarks can be found in Section 4.

2 Materials and methods

2.1 Model selection and development

The term generalised linear model (GLM) refers to a larger class of models which include linear regression, ANOVA, Poisson regression and log-linear models popularised by (McCullagh and Nelder, 1989). GLM describes the patterns of interactions and the model parameters provide measures of strength of associations. In GLM, the dependent variable is assumed to follow an exponential family distribution with mean μ_i , which is assumed to be a nonlinear function of known parameters, but McCullagh and Nelder consider them to be linear, because the known parameters affect the distribution of dependent parameter only through the linear combination.

Dengue is a continuous response variable. However, the relationship between rainfall and dengue is non-linear because heavy rainfall and maximum temperature can severely affect vector abundance downwards. Therefore, we use the generalised linear regression model as a GLM component for this study. The theory of the generalised linear regression model requires data that generally follow a multivariate normal distribution with the same covariance. Let D_t , R_t , MT_t and mT_t denote number of dengue incidence, rainfall, maximum temperature and minimum temperature at time t . In our study t corresponds to the time in weeks. We use *glmfit* package in *MATLAB* to estimate parameters $\beta_0, \beta_1, \beta_2$ and β_3 based on distribution fitting. With the assumption of linearity D_t is expressed as follows.

$$D_t = \beta_0 + \beta_1(R_t) + \beta_2(MT_t) + \beta_3(mT_t) \quad (1)$$

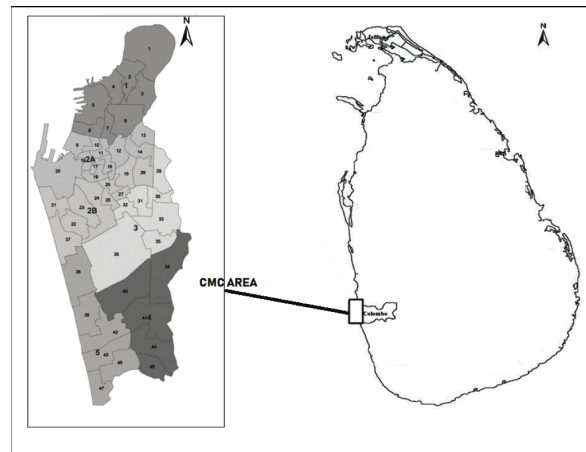
Recall that dengue incidence have strong correlation with rainfall with time lags. Therefore, it is possible to obtain equation (2) by including time lag into rainfall, maximum temperature and minimum temperature variables in equation (1).

$$D_t = \beta_0 + \beta_1(R_{t-lag_R}) + \beta_2(MT_{t-lag_{(MT)}}) + \beta_3(mT_{t-lag_{(mT)}}) \quad (2)$$

2.2 Study area

Now we will describe in detail the significance of CMC area as a ideal geographical territory to evaluate the model. The CMC is the largest city and financial centre in Sri Lanka located in the west cost of the island. Figure 2 demonstrates the boundaries of the area. The CMC area is the most densely populated area in the country where 555031 population living in 37 km² area according to the 2012 Census. Majority of the corporations head offices, factories and a number of prominent public schools are located within the CMC area. Furthermore, public transport service connects Colombo to other major cities in the island. Hence, viral disease can be easily transmitted from Colombo to other cities.

Figure 2 Colombo municipal council area map



CMC area is also influenced by two monsoon seasons, Northeast monsoon season from December to February and Southwest monsoon season from May to September. Moreover, temperature of urban Colombo varies from 22° to 33°. The most important factor is that the CMC area reports the highest numbers of dengue cases every year according to Epidemiology unit, Department of Health data. Several previous researches have been conducted to identify the correlation between meteorological factors and dengue within the area (Goto et al., 2013; Pathirana et al., 2009; Wickramaarachchi and Perera, 2017). Due to the highly complicated transmission dynamics of dengue, existing models failed to predict outbreaks. Accordingly, yielding a better outbreak prediction model for CMC area is vital in controlling the spread of the disease.

2.3 Data collection and analysis

Dengue Fever and DHF are categorised in group B of notifiable disease in Sri Lanka. An online notification system was built in 2012 to collect relevant information of dengue

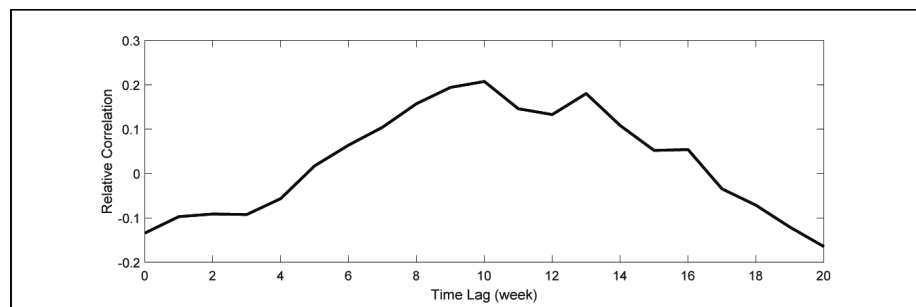
patients from sentinel hospitals in all parts of country. Then information is communicated through weekly Epidemiological report. For our study we have considered weekly reported dengue incidence gained from Epidemiology unit, Department of Health, Sri Lanka, weekly data of rainfall, maximum temperature and minimum temperature data from Meteorological Department, Sri Lanka from 2009 to 2015 for CMC area. For the comparison between the proposed model with currently used method, we have considered weekly reported dengue incidence from 2006 to 2015. In this study, it is needed to examine the lag time between climate factors and dengue incidence. To analyse the correlation between weekly dengue incidence and climate factors Pearson cross correlation formula has been used. Pearson correlation coefficient is the standard coefficient that can be used to measure the strength of the linear relationship between two variables (Hauke and Kossowski, 2011). All analysis and simulations were performed using *MATLAB*.

3 Results and discussion

Considering the life cycle of *Aedes* mosquitoes, it can be noticed that the *Aedes* mosquitoes proceed life cycle from eggs to adult through larvae and pupae and it takes approximately 1–2 weeks or longer depending on temperature, availability of water and nutrients (Berkhout et al., 2013). Another interesting fact of *Aedes* mosquitoes is that the mosquito eggs can resist without drying out for a few months until they have favourable conditions (Sota and Mogi, 1992). Also notice that, the incubation period of infected human ranges from 3 to 14 days. Those factors motivated and thrived us to measure the time delay of dengue infection with rainfall.

In order to calculate the time delay, the Pearson correlation formula has been used and the correlation measure was plotted with time lags from 0 to 20 weeks. Here we assume that the disease occurs within or after rainy seasons. Figure 3 represents the correlation measures between weekly rainfall data and weekly dengue data for the period of 2009 to 2015 in CMC area with time delay.

Figure 3 Correlation of dengue data with rainfall



By Figure 3, it can be observed that the highest correlation between dengue and rainfalls occurs with a 10-week delay. Therefore, we can use value 10 for lag_R in equation (2).

Recall that temperature influences vector biology and vector-virus interactions. For examples, *Aedes* mosquitoes emerge from eggs to adults in a shorter period at higher temperature and the mortality rates of adult mosquitoes increase with increasing temperature

above 30°C (Hii, 2013). Furthermore, infected human experience a shorter incubation period for dengue viruses with high temperature (Berkhout et al., 2013). Those factors motivated us to calculate the time delay of reported dengue incidence with maximum temperature.

By Figure 4, it can be observed that the highest correlation between dengue and maximum temperature occurs with a 16-week delay. Similarly, we examined the time delay of dengue incidence with minimum temperature. Figure 5 represents the correlation values between minimum temperature and dengue incidence against time lag. By Figure 5, it can be observed that the highest correlation occurs with a 13-week delay.

Figure 4 Correlation of dengue data with maximum temperature

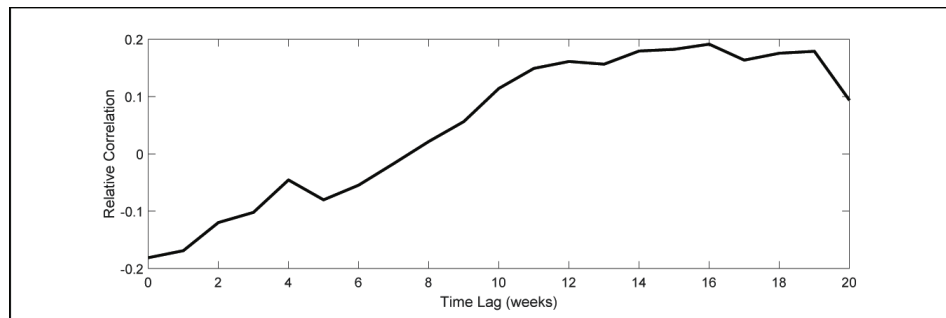
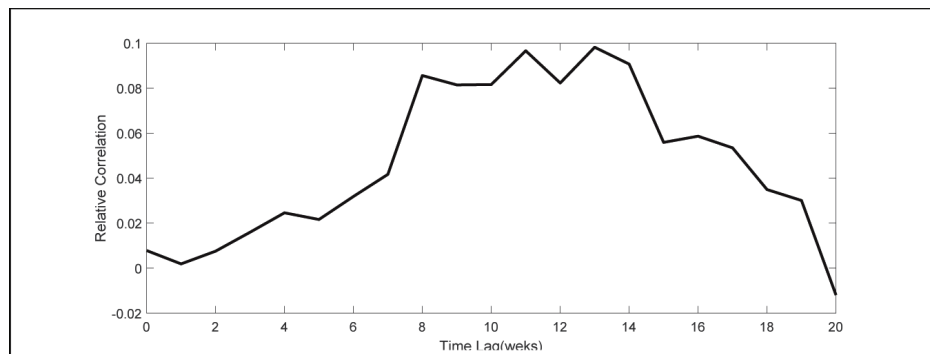


Figure 5 Correlation of dengue data with minimum temperature



According to Figures 3–5 lag times for the highest cross-correlation value between dengue–rainfall, dengue–maximum temperature and dengue–minimum temperature are 10, 16 and 13 respectively. After cross correlation analysis, our next task was to determine the distribution fitting into the dengue inhabitants data. For that purpose we used raw data and log dengue data. Since the dataset of weekly dengue incidence contains zeros, we added one for each data point before calculating the logarithm. To assess the distribution we used *histfit* function in *MATLAB*. Among raw data and log dengue data, best distribution fitting has been given by log dengue data with normal distribution. Since the theory of multiple linear regression models requires data that generally follows a multivariate normal distribution, we used log dengue inhabitants data and Figure 6 represents normal distribution fitting for log dengue data.

In order to capture the outbreaks of the disease in urban Colombo according to climate effects, equation (2) has been changed based on the results of analysed data. Recall the life cycle of *Aedes* mosquitoes takes approximately two weeks and incubation period of infected human ranges from 3 to 14 days, the number of adult mosquitoes at time t depends on the number of adult mosquitoes at time $t - 4$. Since calculating the number of adult mosquitoes is infeasible and infected mosquitoes density depends upon number of infected human population, we included reported dengue data with 4 weeks time delay to the moderated function. Equation (3) represents the moderated formulation.

$$\log(D_t + 1) = \beta_0 + \beta_1(R_{t-10}) + \beta_2(MT_{t-16}) + \beta_3(mT_{t-13}) + \beta_4(D_{t-4}) \quad (3)$$

Figure 7 represents the comparison between simulation results and actual data. We also defined threshold value for dengue incidence which is the average number of reported dengue cases for the time period 2009 to 2015.

Figure 6 Normal distribution fitting for log dengue incidence histogram

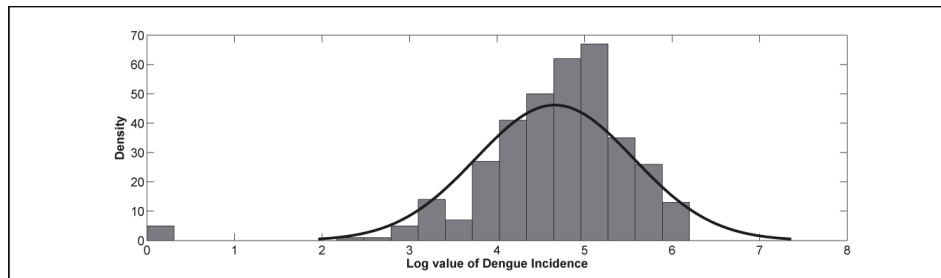
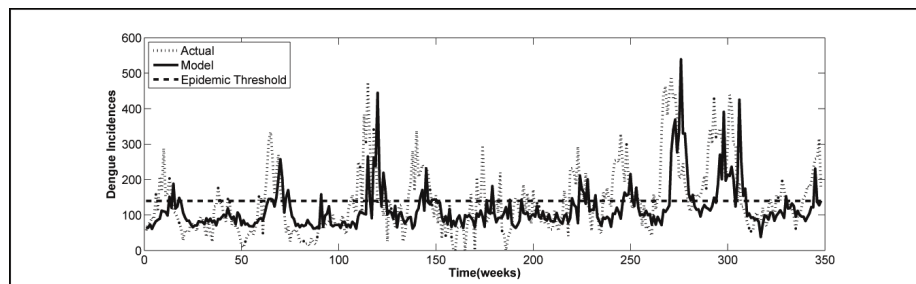


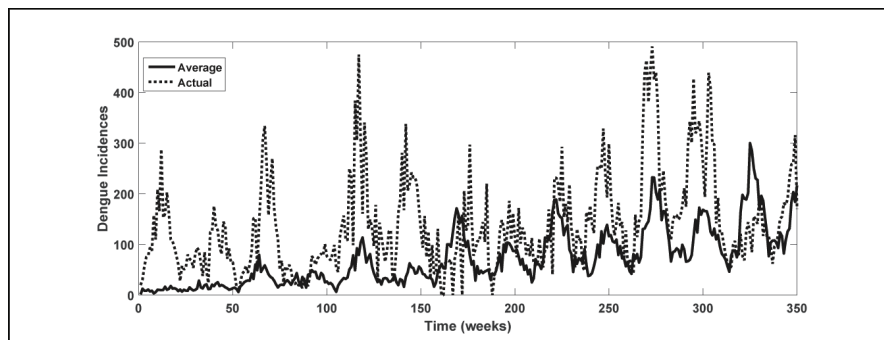
Figure 7 Infected inhabitants population by statistical model



From Figure 7, it can be observed that the simulated data captures the magnitude of the outbreaks. Furthermore, it can be noticed that the threshold value can be used to identify the outbreak situation. According to weekly reports of Epidemiology Unit, Department of Health, Sri Lanka, existing method for outbreak prediction is moving weekly average taken over a period of 5 years. Figure 8 represents the comparison between existing method and actual data. From Figure 8, it can be observed that existing method does not capture the outbreak situations and also the 5 years moving weekly average value cannot be taken as a threshold value to identify the disease outbreak.

Modelling early detection of disease outbreaks has two main advantages. First is identifying the duration and second is predicting the potential outbreak size. For the moderated regression model we used rainfall data, maximum temperature, minimum temperature and dengue data itself with 10, 16, 13 and 4 weeks time lag respectively. Hence, the dynamics of the disease can be predicted one month ahead by the proposed model using real time data. Then the outbreaks of the disease can be identified by comparing the predicted data with the threshold value. Therefore, we can use the proposed model as an real time climate data driven warning system. Provide the dengue outbreak alerts in one month advance is essential for public health authorities to establish efficient vector control strategies. In other words, it provides sufficient time to take control measurement to minimise the spread of the disease.

Figure 8 Comparison of actual infected inhabitants population and average value



Moreover, the proposed model could be used to capture the potential outbreak size. Hence, health authorities could improve health-planning during the outbreak period and it would yield much benefit particularly in countries with limited resources. Although the proposed model captures the outbreak size, from the Figure 7, it can be noticed that there is a disagreement with time period of outbreaks between simulated data and actual data. Hence, we are looking forward to extend the model by defining a functional format to capture the exact time duration for the outbreaks.

4 Conclusion

Dengue is a rapidly spreading disease in the world which has a complex transmission mechanism. Since dengue transmission is being highly affected by climate changes, estimating the spreading pattern and correlation with weather is vital to control the spreading of the disease. Existing classical mathematical and statistical models based on parameters are insensitive to time or space. Thus, existing models do not have the capacity to fully describe the complex transmission dynamics of dengue. Based on reported dengue data and climate data from 2009 to 2015, we estimated spreading pattern of dengue infection for urban Colombo, Sri Lanka using generalised linear regression model.

The correlation between dengue-rainfall, dengue-maximum temperature and dengue-minimum temperature have been analysed using Pearson cross correlation formula. Since the highest correlation value is less than 0.5 for each pair, it would be worthwhile to measure the correlation against time lags from 0 to 20 weeks for each year for a better prediction.

Relating to analysed data moderated generalised linear regression model used to predict the outbreaks in urban Colombo, Sri Lanka. We are looking forward to extend this model to capture the time gap between simulation results and actual data.

Since dengue is a disease with exponential growth, the magnitude of the outbreak size increases with time. Hence, we are looking forward to define a moving threshold function instead of a constant threshold value. Since the spread of dengue also depends on geographical factors and human mobility, it might be an interesting future task to extend the model using these factors.

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